

Asymptotic Analysis Comprehensions

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Figure 1: From order evaluating to asymptopia ¡—

1 Basics of order, $\mathcal{O}$ and o

1.1 Summary/Comprehensions

The knowledge of orders of 2 infinitesimal or infinite quantities and the connection between them are common for us. Here we list some notes you may need to pay attention to:

1. Sometimes the "big $\mathcal{O}$ constant" in a strong function can be related to the argument, and the same is true for the small "o".

Example. $\sin(xy) = \mathcal{O}(1)$, $\sin(xy) = \mathcal{O}_y(x)$

2. The infinitesimal of 2 functions can't always keep for their inverse functions, i.e. $\varphi(x) = o(f(x)) \not\Rightarrow \tilde{f}(x) = o(\tilde{\varphi}(x))$. (Here: $\varphi(x)$ and $f(x)$ are increasing functions, as $\tilde{\varphi}(x)$ and $\tilde{f}(x)$ are their inverse functions)

Example. $f(x) = e^x$, $\varphi(x) = \frac{e^x}{x}$, $x \rightarrow \infty$

3. The elementary of $\mathcal{O}$ and o notation:

Notes. $\varphi = \mathcal{O}(f)$ means an inequality, while $\varphi = o(g)$ express a limitation.

4. 16 Rules of calculating $\mathcal{O}$ and o :

1. Transitivity

$$[1] \lim_{x \rightarrow x_0} f(x) = \infty, \varphi(x) = \mathcal{O}(1) \implies \varphi(x) = o(f(x))$$

$$[2] f = \mathcal{O}(\varphi), \varphi = \mathcal{O}(\psi) \implies f = \mathcal{O}(\psi)$$

$$[3] f = \mathcal{O}(\varphi), \varphi = o(\psi) \implies f = o(\psi)$$

2. $\mathcal{O}$

$$[4] \mathcal{O}(f) + \mathcal{O}(g) = \mathcal{O}(f + g)$$

$$[5] \mathcal{O}(f)\mathcal{O}(g) = \mathcal{O}(fg)$$

3. $\mathcal{O}$ & o

$$[6] o(1)\mathcal{O}(f) = o(f)$$

$$[7] \mathcal{O}(1)o(f) = o(f)$$

$$[8] \mathcal{O}(f) + o(f) = \mathcal{O}(f)$$

4. o

$$[9] o(f) + o(g) = o(|f| + |g|)$$

$$[10] o(f)o(g) = o(fg)$$

5. Power

$$[11] (\mathcal{O}(f))^k = \mathcal{O}_k(f^k), k \in \mathbb{N}$$

$$[12] (o(f))^k = o(f^k)$$

6. Equivalence $\sim$

$$[13] f \sim g, g \sim \varphi \implies f \sim \varphi$$

6. Equivalence $\sim$

$$[14] \quad f = o(g), g \sim \varphi \implies g \sim \varphi \pm kf$$

7. Summation of o

$$[15] \quad f, g > 0, f = o(g), x \rightarrow \infty \implies \int_A^B f(x)dx = o(\int_A^B g(x)dx), \\ B > A \rightarrow \infty$$

$$\text{Specially, if } \exists A_0 \text{ s.t. } \int_{A_0}^{\infty} gdx < \infty, \text{ then } \int_A^{\infty} f(x)dx = o(\int_A^{\infty} g(x)dx), \\ B > A \rightarrow \infty$$

$$[16] \quad a_n, b_n > 0, a_n = o(b_n), n \rightarrow \infty \implies \sum_{n=N}^M a_n = \\ o(\sum_{n=N}^M b_n), M > N \rightarrow \infty$$

$$\text{Specially, if } \sum_{n=1}^{\infty} b_n < \infty, \text{ then } \sum_{n=N}^{\infty} a_n = o(\sum_{n=N}^{\infty} b_n), N \rightarrow \infty$$

5. Extension for Taylor's formula: [/Tech/](#)

Theorem 1.1. *If $f^{(n)}(x)$ exists on $U(x_0; \delta)$ and $|f^{(n)}(x)| \leq M$, then*

$$f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + O(|(x - x_0)^n|)$$

holds on $U(x_0; \delta)$.

Corollary 1.1. 1. $\sin x = x - \frac{x^3}{3!} + O(x^5)$, $x \rightarrow 0$ or $|x| < \infty$

$$2. \cos x = 1 - \frac{x^2}{2!} + O(x^4), x \rightarrow 0 \text{ or } |x| < \infty$$

$$3. \log(1+x) = x - \frac{x^2}{2} + O(x^3), x \rightarrow 0$$

$$4. (1+x)^\alpha = 1 + \alpha x + O(x^2), x \rightarrow 0$$

$$5. e^x = 1 + x + O(x^2), x \rightarrow 0 \text{ or } |x| < 1$$

Notes. 1. Often $\log(1+x) = x + O(x^2)$ is enough to use.

$$2. \tan x = 1 + \frac{x^3}{3} + O(x^5).$$

Corollary 1.2. *If f satisfies $\lim_{x \rightarrow x_0} f(x) = 0$, then*

$$1. \sin f(x) = f(x) - \frac{f(x)^3}{3!} + O(f(x)^5), f(x) \rightarrow 0 \text{ or } |f(x)| < \infty$$

$$2. \cos f(x) = 1 - \frac{f(x)^2}{2!} + O(f(x)^4), f(x) \rightarrow 0 \text{ or } |f(x)| < \infty$$

$$3. \log(1+f(x)) = f(x) - \frac{f(x)^2}{2} + O(f(x)^3), f(x) \rightarrow 0$$

$$4. (1+f(x))^\alpha = 1 + \alpha f(x) + O(f(x)^2), f(x) \rightarrow 0$$

$$5. e^{f(x)} = 1 + f(x) + O(f(x)^2), f(x) \rightarrow 0 \text{ or } |f(x)| < 1$$

6. Abel type theorem: [/Tech/](#)

Theorem 1.2. Given $b_n > 0$, $\sum_{n=1}^{\infty} b_n = \infty$, $a_n = o(b_n)$, $n \rightarrow \infty$, then

$$\sum_{n=1}^N a_n = o\left(\sum_{n=1}^N b_n\right), \quad N \rightarrow \infty.$$

<Abel type Theorem - 1>:
(dicrete) <Stolz's Theorem>

Solution. Denote K_n : the greatest k s.t. $\sum_{i=1}^k b_i < \sqrt{\sum_{i=1}^n b_i}$ holds, then $K_n \rightarrow \infty$, $n \rightarrow \infty$.
Then $|\sum_{i=1}^n a_i| = O(1) \sum_{i=1}^{K_n} b_i + o(1) \sum_{i=K_n}^n b_i = O(1) \sqrt{\sum_{i=1}^n b_i} + o(1) \sum_{i=1}^n b_i = o(1) \sum_{i=1}^n b_i$.

Notes. 1. Comparing with ordinary $\epsilon - N$ method, the oder estimation proof method is concise and elegant.
2. The commutive of $\sum$ and o .

Theorem 1.3. Given $g(x) > 0$, $\int_a^{\infty} g(x)dx$, $f(x) = o(g(x))$, $x \rightarrow \infty$, then

$$\int_a^x f(t)dt = o\left(\int_a^x g(t)dt\right), \quad x \rightarrow \infty.$$

(continue)
<L'Hospital Theorem>

Solution. Proof idea is same with dicrete form.

Theorem 1.4. Given $b_n > 0$, $a_n = o(b_n)$, $f(x) = \sum_{n=0}^{\infty} a_n x^n$, $g(x) = \sum_{n=0}^{\infty} b_n x^n$, and $g(x) = \begin{cases} < \infty & 0 \leq x < 1, \\ \rightarrow \infty & x \rightarrow 1^- \end{cases}$, then

<Abel type Theorem - 2>
<Abel's Limits Theorem>

$$f(x) = o(g(x)), \quad x \rightarrow 1^-.$$

Corollary 1.3. Given $f(x)$ and $g(x)$ satisfying the conditions of Thm 1.4, and $a_n = b_n(1 + o(1))$, $n \rightarrow \infty$, then

$$f(x) = g(x)(1 + o(1)), \quad x \rightarrow 1^-.$$

Solution. Proof idea: consider $c_n = a_n - b_n$ and $f(x) - g(x)$.

Corollary 1.4. Given $f(x)$ and $g(x)$ satisfying the conditions of Thm 1.4, and converging on $[0, 1)$. Let $s_n = \sum_{k=0}^n a_k$ and $t_n = \sum_{k=0}^n b_k$ satisfy: (1) $t_n > 0$; (2) $\sum_{n=1}^{\infty} t_n x^n \rightarrow \infty$, $x \rightarrow 1^-$; (3) $s_n = o(t_n)$, $n \rightarrow \infty$, then

<Abel type Theorem - 3>
(Weak version of AtT - 2)

$$f(x) = o(g(x)), \quad x \rightarrow 1^-.$$

Solution. /Tech tool/: Power Series Multiplication Theorem
 $F(x) = \frac{1}{1-x} f(x) = \sum_{n=0}^{\infty} s_n x^n$, $G(x) = \frac{1}{1-x} g(x) = \sum_{n=0}^{\infty} t_n x^n$, $0 \leq x < 1$.

Corollary 1.5. Given $f(x)$ and $g(x)$ satisfying the conditions of Thm 1.4, and converging on $[0, 1)$. Let $s_n = \sum_{k=0}^n a_k$ and $t_n = \sum_{k=0}^n b_k$ satisfy: (1) $t_n > 0$; (2) $\sum_{n=1}^{\infty} t_n x^n \rightarrow \infty$, $x \rightarrow 1^-$; (3) $s_n \sim t_n$, $n \rightarrow \infty$, then

(Weak version of AtT - 2)

$$f(x) \sim g(x), \quad x \rightarrow 1^-.$$

7. Γ -function and Stirling's Formula.

Definition 1.1. Given $x \in \mathbb{C}$, $\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$, $\operatorname{Re}(x) > 0$.

5 properties of $\Gamma(x)$:

1. $\Gamma(x) = (x-1)\Gamma(x-1) = \cdots = (x-1)\cdots(x-n)\Gamma(x-n)$
Specially, $\Gamma(n+1) = n!$.

2. [/Tech tool/](#)

Lemma 1. $\forall \alpha, \beta > 0$, $\frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)} = \int_0^\infty \frac{t^{\beta-1}}{(1+t)^{\alpha+\beta}} dt = \int_0^1 y^{\alpha-1}(1-y)^{\beta-1} dy$.

Pre-Lemma 1. If $\forall b > a$, we have

$$\int_a^b dx \int_\alpha^\beta f(x, y) dy = \int_\alpha^\beta dy \int_a^b f(x, y) dx,$$

where $a, \beta < \infty$, all are proper integrals; and if $\int_a^\infty f(x, y) dx$ converges uniformly for $\alpha \leq y \leq \beta$, then

$$\int_a^\infty dx \int_\alpha^\beta f(x, y) dy = \int_\alpha^\beta dy \int_a^\infty f(x, y) dx.$$

Pre-Lemma 2. Suppose $f(x, y) \geq 0$. If $\forall c < b$ and $\gamma < \beta$, we have

$$\int_a^c dx \int_\alpha^\beta f(x, y) dy = \int_\alpha^\beta dy \int_a^c f(x, y) dx$$

and

$$\int_a^b dx \int_\alpha^\gamma f(x, y) dy = \int_\alpha^\gamma dy \int_a^b f(x, y) dx,$$

then when one of the following expressions exists, we have

$$\int_a^b dx \int_\alpha^\beta f(x, y) dy = \int_\alpha^\beta dy \int_a^b f(x, y) dx;$$

when b or β is infinite, the above conclusion still holds.

3. [/Tech tool/](#) (Legendre duplication formula)

Lemma 2. $\Gamma(2x)\Gamma\left(\frac{1}{2}\right) = 2^{2x-1}\Gamma(x)\Gamma\left(x + \frac{1}{2}\right)$, where $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.

4. [/Tech tool/](#)

Lemma 3. $\exists C$, s.t. $\log \Gamma(n) = (n - \frac{1}{2}) \log n - n + C + O\left(\frac{1}{n}\right)$.

5. [/Tech tool/](#)

Lemma 4. $\frac{\Gamma(x)}{\Gamma(x+a)} = x^{-a} + O(x^{-a-1})$.

The proof of some important of them will be displayed at the end of this chapter, hope you could try to feel and understand them, or even prove them by

yourself.

Theorem 1.5. [/Tech/](#)

$$\Gamma(x) = x^{x-\frac{1}{2}} e^{-x} \sqrt{2\pi} (1 + O(\frac{1}{x})), \quad x > 0.$$

Solution. Just consider $x \in \mathbb{R}/\mathbb{Z}$, since $\Gamma(x) \in C[0, \infty)$. Let $x = n + a$, where $n \in \mathbb{Z}, 0 < a < 1$, then

$$\begin{aligned} \log \Gamma(x) &= \log \Gamma(n + a) = \log \Gamma(n) + a \log n + O(\frac{1}{n}) \quad (\text{Lemma 4}) \\ &= (n - \frac{1}{2}) \log n - n + C + a \log n + O(\frac{1}{n}) \quad (\text{Lemma 3}) \\ &= (x - a - \frac{1}{2}) \log(x - a) - x + a + C + a \log(x - a) + O(\frac{1}{x}) \\ &= [x' \triangleq x - a] = (x - \frac{1}{2}) \log x - x + C + O(\frac{1}{x}) \quad (1) \end{aligned}$$

Also $\log \Gamma(2x) + \log \Gamma(\frac{1}{2}) = (2x - 1) \log 2 + \log \Gamma(x) + \log \Gamma(x + \frac{1}{2})$ (Lemma 2), take the $\log \Gamma(x)$ expression (1) into LHS:

$$\begin{aligned} (2x - \frac{1}{2}) \log(2x) - 2x + C + \log \Gamma(\frac{1}{2}) + O(\frac{1}{x}) \\ = (2x - 1) \log 2 + (x - \frac{1}{2}) \log x + x \log(x + \frac{1}{2}) \\ - 2x - \frac{1}{2} + 2C + O(\frac{1}{x}) \end{aligned}$$

Equating the constant terms on both sides, we get $C = \log \sqrt{2\pi}$, substituting into (1), we obtain the conclusion of the theorem.

Corollary 1.6. (*Stirling's Formula*) [/Tech/](#)

$$n! = n^{n+\frac{1}{2}} e^{-n} \sqrt{2\pi} (1 + O(\frac{1}{n})) = \sqrt{2\pi n} (\frac{n}{e})^n (1 + O(\frac{1}{n})).$$

Solution. $n! = \Gamma(n + 1) = (n + 1)^{n+\frac{1}{2}} e^{-(n+1)} \sqrt{2\pi} (1 + O(\frac{1}{n}))$ (Thm 1.5) and

$$\begin{aligned} (n + 1)^{n+\frac{1}{2}} &= n^{n+\frac{1}{2}} (1 + \frac{1}{n})^{n+\frac{1}{2}} = n^{n+\frac{1}{2}} (1 + \frac{1}{n})^n (1 + O(\frac{1}{n})) \\ &= n^{n+\frac{1}{2}} e^{n \log(1+\frac{1}{n})} (1 + O(\frac{1}{n})) \\ &= n^{n+\frac{1}{2}} e^{n(\frac{1}{n} + O(\frac{1}{n^2}))} (1 + O(\frac{1}{n})) \\ &= n^{n+\frac{1}{2}} e^{1+O(\frac{1}{n})} (1 + O(\frac{1}{n})) = n^{n+\frac{1}{2}} e (1 + O(\frac{1}{n})) \end{aligned}$$

Notes. 1. Observing Stirling's Formula and previous estimation experience, we just need to estimate part $(n+1)^{n+\frac{1}{2}} = n^{n+\frac{1}{2}}(1+\frac{1}{n})^{n+\frac{1}{2}}$. (This kind of trick will be encountered many times.)

2. Tips: 1°. $(1+O(\frac{1}{n})+O(\frac{1}{n^2})) = 1+O(\frac{1}{n})$; 2°. $(1+O(\frac{1}{n}))^k = 1+O(\frac{1}{n})$, $k \in \mathbb{Z}^+$;
3°. $O(\frac{1}{n}) = o(1), n \rightarrow \infty$.

1.2 Problem Sets

Problem. (Infinitesimal calculation) $\int_{x-2}^{\infty} e^{-t} \log t \, dt = o(1)$, $x \rightarrow 0$.

$$\text{Solution. } \int_{x-2}^{\infty} e^{-t} \log t \, dt < \int_{x-2}^{\infty} e^{-\frac{t}{2}} \, dt = -2e^{-\frac{t}{2}} \Big|_{x-2}^{\infty} = 2e^{-\frac{1}{2x^2}}$$

Problem.

(Classical limit review) $x^A = o((1+a)^{\epsilon x})$, $x \rightarrow \infty$, as $\epsilon, a > 0 \, \forall A$.

(Develop) $(\log(x))^A = o(x^{\epsilon})$, $x \rightarrow \infty$,

$(f(x))^A = o(e^{\epsilon f(x)})$, $x \rightarrow \infty$.

$$\begin{aligned} \text{Solution. Let } n &\triangleq [x], \, m \triangleq [A] + 1, \text{ then } (1+a)^x \geq (1+a)^n \geq \\ C_n^{m+1} a^{m+1} (x \rightarrow \infty, \text{ i.e. } n \rightarrow \infty, \text{ and } n > 2m+1) &\geq \frac{(n-m)^{m+1}}{(m+1)!} a^{m+1} \geq \\ \frac{n^{m+1}}{2^{m+1}(m+1)!} a^{m+1}, \text{ then } \frac{(1+a)^x}{x^A} &\geq \frac{n^{m+1} a^{m+1}}{2^{m+1}(m+1)!(n+1)^m} \geq \frac{na^{m+1}}{2^{2m+1}(m+1)!}, \text{ then} \\ \lim_{x \rightarrow \infty} \frac{x^A}{(1+a)^x} &= 0, \text{ i.e. } x^A = o((1+a)^x), \, x \rightarrow \infty. \text{ Since } \lim_{x \rightarrow \infty} \frac{x^A}{(1+a)^{\epsilon x}} = \\ \lim_{x \rightarrow \infty} \left(\frac{x^{\frac{A}{\epsilon}}}{(1+a)^x} \right)^{\epsilon} &= 0, \text{ then } x^A = o((1+a)^{\epsilon x}), \, x \rightarrow \infty. \end{aligned}$$

Problem. /Tech tool/

Let $a_n = O(b_n)$, $n \geq 1$, and series $\sum_{n=1}^{\infty} b_n < \infty$, then $\exists C$, s.t. $\sum_{k=1}^n a_k = C + O(\sum_{k=n+1}^{\infty} b_k)$.

$$\text{Solution. Obviously, series } \sum_{k=1}^{\infty} a_k \text{ converges, then } \exists C \text{ s.t. } \sum_{k=1}^{\infty} a_k = C, \text{ then } \sum_{k=1}^n a_k = \sum_{k=1}^{\infty} a_k - \sum_{k=n+1}^{\infty} a_k = C + O(\sum_{k=n+1}^{\infty} b_k).$$

Problem. /Tech tip/

The 1st principle of order evaluating (specially need using Taylor's series to evaluate) is paying more attention on and catching the infinitesimal or infinite quantities which those are not so good to distinguish directly "by eyes".

Here are some useful techs and interesting examples:

1. /Tech tool/ $\sqrt[n]{n} = e^{\frac{\log n}{n}} = 1 + \frac{\log n}{n} + O(\frac{\log^2 n}{n^2})$, $n \rightarrow \infty$
2. Using 1, you could get $(n+a)^{n+b} = n^{n+b} e^a (1 + \frac{a(b-\frac{a}{2})}{n} + O(\frac{1}{n^2})) \implies (1+\frac{a}{n})^{n+b} = e^a (1 + \frac{a(b-\frac{a}{2})}{n} + O(\frac{1}{n^2}))$ (development of $\lim_{n \rightarrow \infty} (1+\frac{1}{n})^n = e$).
3. Layer-by-layer (from the outside in)
$$\begin{aligned} (x + (x + (x^{\frac{\alpha}{2}}))^{\frac{1}{2}})^{\frac{1}{2}} &= x^{\frac{1}{2}} (1 + (x^{-1} + x^{\frac{\alpha}{2}-2})^{\frac{1}{2}})^{\frac{1}{2}} \quad (0 < \alpha < 2) \\ &= x^{\frac{1}{2}} (1 + \frac{1}{2}(x^{-1} + x^{\frac{\alpha}{2}-2})^{\frac{1}{2}} + O(x^{-1})) \\ &= x^{\frac{1}{2}} (1 + \frac{1}{2}x^{-\frac{1}{2}}(1 + x^{\frac{\alpha}{2}-1})^{\frac{1}{2}} + O(x^{-1})) \\ &= x^{\frac{1}{2}} + \frac{1}{2}(1 + \frac{1}{2}x^{\frac{\alpha}{2}-1} + O(x^{\alpha-2})) + O(x^{-1}) \end{aligned}$$

4. $\lim_{n \rightarrow \infty} \cos^n \frac{x}{\sqrt{n}} = \lim_{n \rightarrow \infty} e^{n \log(1 - \frac{x^2}{2n} + O_x(\frac{1}{n^2}))} = e^{-\frac{x^2}{2}} (1 + O_x(\frac{1}{n}))$
 Related (just form): Wallis' integral $W_n = \int_0^{\frac{\pi}{2}} (\cos x)^n dx = \int_0^{\frac{\pi}{2}} (\sin x)^n dx = \frac{n-1}{n} W_{n-2} = \frac{(n-1)!!}{n!!} (\frac{\pi}{2})^{\delta_{(n \bmod 2)0}} = \sqrt{\frac{\pi}{2n}} (1 + o(1)),$
 $(n \rightarrow \infty)$
5. Using [15], [16] of 16 rules, we could test convergence of some infinite integral or series:
 (1) $\int_1^\infty \frac{(e^{\frac{1}{2x}} - 1)^a}{\log^b(1 + \frac{1}{x})} dx$; (2) $\sum_{n=1}^\infty \frac{1}{\log(1+n)} \sin \frac{1}{n^\beta}$, $\beta > 0$; (3) $\int_1^\infty \frac{\log(1 + \sin \frac{1}{x^\alpha})}{x^\beta \log \cos \frac{1}{x}} dx$, $\alpha > 0$; (4) $\sum_{n=1}^\infty \frac{(1 - \cos \frac{1}{n})^\beta}{\log^\gamma(1+n)}$; (5) $\int_1^\infty \frac{\log(1 + \frac{1}{x^p})}{x^p \sqrt[p]{1+x^2}} dx$, $p \in \mathbb{R}$; (6) $\int_1^\infty \frac{x^m \arctan x}{2+x^n} dx$, $n > 0$; (7) $\int_0^1 \frac{\log x}{(1-x^2)^a (\sin x)^p} dx$, $p \in \mathbb{R}$

Problem. /Basic/ /Tech/

Given $f(x) = o(1)$, $g(x) = o(1)$, $b \neq 0$, then

$$\frac{a+o(1)}{b+o(1)} = \frac{a}{b} + o(1)$$

$$\begin{aligned} \frac{a+f(x)}{b+g(x)} &= \frac{a+f(x)}{b} \left(1 + \frac{g(x)}{b}\right)^{-1} = \frac{a+f(x)}{b} \left(1 - \frac{g(x)}{b} + O(g^2(x))\right) \\ &= \frac{a}{b} + \frac{f(x)}{b} - \frac{ag(x)}{b^2} + O(f(x)g(x)) + O(g^2(x)). \end{aligned}$$

Problem. Applications of Abel type Theorem.

1. Given $y_{n+1} > y_n$, $y_n \rightarrow \infty$, then $\lim_{n \rightarrow \infty} \frac{x_n}{y_n} = \lim_{n \rightarrow \infty} \frac{x_n - x_{n-1}}{y_n - y_{n-1}}$. (Assume LHS limit exists.)
2. Given $a_n \sim \log n$, $n \rightarrow \infty$, then $\sum_{n=0}^\infty a_n x^n \sim \frac{1}{1-x} \log \frac{1}{1-x}$, $x \rightarrow 1^-$.

Solution. 1. Let $\lim_{n \rightarrow \infty} \frac{x_n - x_{n-1}}{y_n - y_{n-1}} = a < \infty$ (If LHS limit is ∞ , then consider LHS $^{-1}$), using Abel type Theorem - 1 (discrete):

$$\begin{aligned} x_k - x_{k-1} &= a(y_k - y_{k-1}) + o(y_k - y_{k-1}), \\ x_n - x_1 &= \sum_{k=2}^n (x_k - x_{k-1}) = a \sum_{k=2}^n (y_k - y_{k-1}) + \sum_{k=2}^n o(y_k - y_{k-1}) \\ &= a(y_n - y_1) + o\left(\sum_{k=2}^n (y_k - y_{k-1})\right) = a(y_n - y_1) + o(y_n) \end{aligned}$$

2. Due to $\frac{1}{1-x} \log \frac{1}{1-x} = \sum_{n=0}^\infty x^n \sum_{n=1}^\infty \frac{x^n}{n} = \sum_{n=0}^\infty \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}\right) x^n$ and $\sum_{k=1}^n \frac{1}{k} \sim \log n$, use Abel type Theorem - 2 (Limits) Corollary 1.

2 Appendix

2.1 Extending tests for convergence of number series

2.1.1 Talking from Cauchy Condensation Test

Theorem 2.1. (*Cauchy Condensation Test*) [Tech toolkit](#)
 $\{a_n\}_n^\infty \searrow^+$ (i.e. $\dots \geq a_n \geq a_{n-1} \geq \dots \geq a_1 \geq 0$), then $\sum_{n=1}^\infty a_n$ and $\sum_{k=1}^\infty 2^k a_{2^k}$ have same convergence.

Connections between Condensation test and Integral test:

$$\begin{aligned} \sum_{n=1}^\infty a_n < \infty &\iff \int_1^\infty f(x)dx < \infty \text{ substitute with: } x = 2^y \\ &\iff \int_1^\infty 2^y f(2^y)dy < \infty \iff \sum_{k=1}^\infty 2^k a_{2^k} < \infty \end{aligned}$$

Theorem 2.2. (*Maclaurin-Cauchy Integral Test*) [Tech toolkit](#)
 $f \searrow^+$ on $[1, +\infty)$, then $\sum_{n=1}^\infty f(n)$ and $\int_1^\infty f(x)dx$ have same convergence.

Here are some examples we could use Condensation test to solve:

1. $\sum_{n=1}^\infty \frac{1}{n^p}$, $\sum_{n=2}^\infty \frac{1}{n \log^p n}$, $\sum_{n=2}^\infty \frac{1}{(\log n)^p}$ (combine Cauchy root test),
 $\sum_{n=2}^\infty \frac{\log n}{n^2}$
2. $\sum_{n=16}^\infty \frac{1}{n(\log n)(\log \log n)(\log \log \log n)^p}$
3. More general, let's consider $f(x) := n^{-a}(\log n)^{-b}(\log \log n)^{-c}$,
 $\sum f(n) < \infty$, $a > 1$, when $a = 1$ condensation transformation gives
 $\sum n^{-b}(\log n)^{-c}$ (seems like the logarithms "shift to the left") so when
 $a = 1$, it converges as $b > 1$. When $b = 1$, the value of c enters.
This result readily generalizes: the condensation test, applied repeatedly,
can be used to show that for $k = 1, 2, 3, \dots$, the generalized
Bertrand series $\sum_{n \geq N} \frac{1}{n \cdot \log n \cdot \log \log n \cdots \log^{o(k-1)} n \cdot (\log^{o^k} n)^\alpha}$ ($N =$
 $\lfloor \exp^{o^k}(0) \rfloor + 1$) converges for $\alpha > 1$ and diverges for $0 \leq \alpha \leq 1$.
from Wikipedia: Cauchy Condensation Test

Besides, the Cauchy condensation test is a partial case of following classical result deu to Schlomilch:

Theorem 2.3. (*Schlomilch*)
 $\{a_n\}_n^\infty \searrow^+$, $\{u_n\}_0^\infty : u_0 < u_1 < \dots$ and $\frac{\Delta u_n}{\Delta u_{n-1}} \leq C$, then $\sum_{n=1}^\infty a_n$ and $\sum_{n=1}^\infty \Delta u_n a_{u_n} = \sum_{n=1}^\infty (u_{n+1} - u_n) a_{u_n}$ have same convergence.

Notes. 1. You may note it that let $u_n = 2^n$, it's Cauchy condensation test. Moreover, you could also use $u_n = p^n$, $p > 1$.

2.1.2 History of tests with monotonicity assumption for sequence

Let's recall some convergence tests with monotonicity assumption for sequence:

Theorem 2.4. (*Ermakov*)

Given $f \searrow_{\geq 0} \in C(1, \infty)$, if $\lim_{t \rightarrow \infty} \frac{e^t f(e^t)}{f(t)} \leq q < 1$, then $\sum_{n=1}^{\infty} f(n) < \infty$;
if $\lim_{t \rightarrow \infty} \frac{e^t f(e^t)}{f(t)} \geq 1$, then $\sum_{n=1}^{\infty} f(n)$ diverges.

Notes. 1. You could replace e^t with an arbitrary $\varphi(t) \nearrow_{\geq 0}$ to get a family of tests.

Theorem 2.5. (*Abel-Olivier*)

Given $\{a_n\}_n^\infty : a_n \searrow_{\geq 0} \rightarrow 0$ (monotone null), if $\sum_{n=1}^{\infty} a_n < \infty$, then $na_n = o(1)$, $n \rightarrow \infty$.

Notes. 1. Proof idea: Cauchy convergence criterion.
2. The inverse is not true: $a_n = \frac{1}{n \log n}$.

Theorem 2.6. (*Sapogov*)

Given $\{a_n\}_n^\infty : a_n \searrow_{\geq 0}$, then $\lim_{n \rightarrow \infty} a_n = 0$ i.f.f. $\sum_{n=1}^{\infty} (1 - \frac{a_n}{a_{n+1}})$ as well as $\sum_{n=1}^{\infty} (\frac{a_{n+1}}{a_n} - 1)$ diverges.

Notes. 1. Proof idea: $\sum_{n=1}^{\infty} b_n$ and $\sum_{n=1}^{\infty} \log(1 - b_n)$ has same convergence.
2. It's a simple test for n -th term of a series approach to 0. 3. Other Description:

Given $\{a_n\}_n^\infty : a_n \nearrow_{\geq 0}$, if $\sum_{n=1}^{\infty} (1 - \frac{a_n}{a_{n+1}})$ as well as $\sum_{n=1}^{\infty} (\frac{a_{n+1}}{a_n} - 1)$ converges, then a_n is bounded or diverges otherwise.

Theorem 2.7. (*Dedekind-du Bois Reymond*)

Def: $\{a_n\}_n^\infty \in BV(\text{bounded variation}) : \sum_{n=1}^{\infty} |a_{n+1} - a_n| < \infty$.

1. Given $\{a_n\}_n^\infty \in BV$ also $a_n = o(1)$ and $\sum_{k=1}^n b_k < C$, $\forall n \in \mathbb{N}$, then $\sum_{n=1}^{\infty} a_n b_n < \infty$.

2. Given $\{a_n\}_n^\infty \in BV$ and $\sum_{n=1}^{\infty} b_n < \infty$, then $\sum_{n=1}^{\infty} a_n b_n < \infty$.

Corollary 2.1. 1. (*Dirichlet*)

Given a_n is monotone null and $\sum_{k=1}^n b_k < C$, $\forall n \in \mathbb{N}$, then $\sum_{n=1}^{\infty} a_n b_n < \infty$.

Corollary. (*Leibniz*)

Given a_n is monotone null, then $\sum_{n=1}^{\infty} (-1)^n a_n < \infty$.

2. (*Abel*)

Given a_n is bounded monotone and $\sum_{n=1}^{\infty} b_n < \infty$, then $\sum_{n=1}^{\infty} a_n b_n < \infty$.

Before our formal work, let's introduce some basic more general monotonicity definitions:

Definition 2.1. We call a non-negative null (that is, tending to zero at infinity) sequence a_k weak monotone, written **WMS**, if for some positive absolute constant C it satisfies $a_k \leq C a_n \forall k \in [n, 2n]$.

To introduce a counterpart for functions, we will assume all functions to be

defined on $(0, \infty)$, locally of bounded variation, and vanishing at infinity.

Definition 2.2. We say that a non-negative function f defined on $(0, \infty)$, is weak monotone, written **WM**, if $f(t) \leq Cf(x) \forall t \in [x, 2x]$.

Denote $\Delta a_k = a_{k+1} - a_k$.

Definition 2.3. A positive null sequence a_k is called general monotone, written **GMS**, if it satisfies $\sum_{k=n}^{2n} |\Delta a_k| \leq Ca_n$, for any n and some absolute constant C .

Definition 2.4. We say that a non-negative function f is general monotone, **GM**, if $\forall x \in (0, \infty)$, $\int_x^{2x} |df(t)| \leq Cf(x)$.

- Notes.** 1. The class of quasi-monotone sequences, that is, a_k such that there exists $\tau > 0$ so that $k^{-\tau} a_k \downarrow$, is a proper subclass of *GMS*.
2. One of the simple basic properties of *GMS* is $GMS \subsetneq WMS$.
3. We just remark that if $f(\cdot)$ is a *GM* function, then for $a_k = f(k)$ there holds $a_k \in GMS$.
4. Also $GM \subsetneq WM$.

2.1.3 Extension of the "monotone" tests

Now, let's extend those aforementioned "monotone" tests for sequences of numbers and functions:

Theorem 2.8. (*Maclaurin-Cauchy* $\uparrow$)
Let f be a *WM* function. Then the series $\sum_{k=1}^{\infty} f(k)$ and integral $\int_1^{\infty} f(t)dt$ converge or diverge simultaneously.

Theorem 2.9. (*Ermakov* $\uparrow$)
Let f be a *WM* function and let $\varphi(t)$ be a monotone increasing, positive function having a continuous derivative and satisfying $\varphi(t) > t$ for all t large enough.
If for t large enough $\frac{f(\varphi(t))\varphi'(t)}{f(t)} \leq q < 1$, then series $\sum_{n=1}^{\infty} f(n)$ converges, while if $\frac{f(\varphi(t))\varphi'(t)}{f(t)} \geq 1$, then series $\sum_{n=1}^{\infty} f(n)$ diverges.

Theorem 2.10. (*Schlömilch* $\uparrow$)
Let u_k be an increasing sequence of positive numbers such that $u_{k+1} = O(u_k)$ and $u_k \rightarrow \infty$.
Let f be a *WM* function. Then both series $\sum_{k=1}^{\infty} f(u_k)\Delta u_k$ and $\sum_{k=1}^{\infty} f(u_{k+1})\Delta u_k$ converge (or diverge) with $\int_1^{\infty} f(t)dt$.
Let a_k be a *WMS*. Then $\sum_{n=1}^{\infty} a_n$ converges if and only if the series $\sum_{k=1}^{\infty} \Delta u_k a_{u_k} = \sum_{k=1}^{\infty} (u_{k+1} - u_k) a_{u_k}$ converges.

Theorem 2.11. (*Abel-Olivier* $\uparrow$)
Let a_k be a *WMS*. If series $\sum_{n=1}^{\infty} a_n$ converges, then ka_k is a null sequence.

Analyzing one Dvoretzky's result and its proof, we obtain the following statement:

Theorem 2.12. If $a_k, b_k \in WMS$, and the series are convergent and divergent respectively, then $\forall M > 1$ there $\exists$ infinitely many R_j , $R_j \rightarrow \infty$, s.t. $\forall k$ with $R_j \leq k \leq MR_j$, we have $a_k < b_k$.

Our next result is "dual" to the above Schlömlich-type extensions.

Definition 2.5. An increasing sequence u_k is called **lacunary** if $u_{k+1}/u_k \geq q > 1$.

A more general class of sequences is the one in which each sequence can be split into finitely-many lacunary sequences, which we will write $u_k \in \Lambda$. (This is true i.f.f. $\sum_{j=1}^k u_j \leq Cu_k$.)

In different terms, $u_k \in \Lambda$ is true i.f.f. there $\exists r \in \mathbb{N}$ s.t. $\frac{u_{k+r}}{u_k} \geq q > 1$, $k \in \mathbb{N}$.

Definition 2.6. $\Delta a_{u_k} := a_{u_k} - a_{u_{k+1}}$

Theorem 2.13. Let a_k be a non-negative WMS, and let a sequence u_k be s.t. $u_k \in \Lambda$ and $u_{k+1} = O(u_k)$. Then $\sum_{k=1}^{\infty} a_k$, $\sum_{k=1}^{\infty} u_k |\Delta a_{u_k}|$, and $\sum_{k=1}^{\infty} u_k a_{u_k}$ converge or diverge simultaneously.

It's also possible to get equiconvergence results for an important case $u_n = n$, where the lacunarity is not suitable. (we need to proceed a smaller class than WMS)

Theorem 2.14. Let a_k be a GMS. Then series $\sum_{n=1}^{\infty} a_n$ and $\sum_k k |\Delta a_k|$ converge or diverge simultaneously.

Theorem 2.15. Let f be a GM function. Then the integrals $\int_1^{\infty} f(t)dt$ and $\int_1^{\infty} t |df(t)|$ converge or diverge simultaneously.

Notes. Negative type results:

1. Sapogov type test cannot be true if b_k (as well as $1/b_k$) is WMS. (A counterexample can be constructed against generalization of Abel's test.)
2. As for extending the Leibniz test, it cannot hold without additional assumption of the boundedness of variation: just take $a_k = 1/\ln k$ everywhere except $n = 2^k$ where $a_n = 2/\ln n$.
3. Is there exists a wider class that those tests could hold? (see below problem)

Problem. WMS is, in a sense, the widest class for which such tests are still valid.

We give some examples to show you the problem is seemly true.

Example. 1. An immediate natural extension of WMS is the class defined by $a_k \leq C \sum_{n=k/2}^k \frac{a_n}{n}$, or a bit more general $a_k \leq C \sum_{n=[k/c]}^{[ck]} \frac{a_n}{n}$, $c > 1$. The principle difference between WMS and these classes is that the latter two allow certain amount of zero members, unlike WMS that forbid even a single zero, i.e., $a_{n_0} = 0$ implies $a_n = 0$ for $n \geq n_0$. Putting zeros on certain positions, say $k = 2^n$, we easily construct a counterexample to show that the Cauchy condensation test cannot be

valid for these classes nor its extensions.

In a similar way, just letting a function f to take non-zero values only close to integer points, one sees that the Maclaurin-Cauchy integral test may fail as well.

2. In conclusion, note that WMS is a subclass of the broadly used Δ_2 -class: for which the doubling condition $a_{2k} \leq Ca_k$ holds $\forall k \in \mathbb{N}$.

We observe that assuming the doubling condition by no means can guarantee the above tests to be extended:

To illustrate this, the easiest way is to consider Cauchy's condensation test. Take a_k such that $a_k = \begin{cases} 2^{-k}, & k \neq 2^n; \\ n^{-2}, & k = 2^n. \end{cases}$ This sequence is doubling but not WMS . Obviously, $\sum_{n=1}^{\infty} 2^n a_{2^n} = \sum_{n=1}^{\infty} \frac{2^n}{n^2}$ diverges.

2.1.4 Summary

The main results may be summarized as the following statement:

Theorem 2.16. *Let $f(\cdot) \in WM$. Then the following series and integrals converge or diverge simultaneously:*

$$\int_1^{\infty} f(t)dt, \quad \sum_k f(k);$$

$$\sum_k (u_{k+1} - u_k)f(u_k), \quad \text{provided } u_k \uparrow, \quad u_{k+1} = O(u_k);$$

$$\sum_k u_k f(u_k), \quad \sum_k u_k |f(u_{k+1}) - f(u_k)|, \quad \text{with } u_k \text{ lacunary, } u_{k+1} = O(u_k);$$

$$\sum_k k |f(k+1) - f(k)|, \quad \int_1^{\infty} t |df(t)|, \quad \text{provided } f(\cdot) \text{ is GM.}$$

Example. Besov space:

For $k > r$, $\theta, r > 0$ and $p \geq 1$ $\|f\|_{B_{p,\theta}^r} = \left(\int_0^1 t^{-r\theta-1} \omega_k^\theta(f; t)_p \frac{dt}{t} \right)^{1/\theta}$.

Equivalent form: $\|f\|_{B_{p,\theta}^r} = \left(\int_1^{\infty} t^{r\theta+1} \omega_k^\theta(f; 1/t)_p \frac{dt}{t} \right)^{1/\theta}$.

Since $\omega_k(f; t)_p \uparrow$, the function under the integral sign is WM . Applying generalized Maclaurin-Cauchy and Cauchy condensation tests, we obtain convenient equivalent form: $\left(\sum_{\nu=0}^{\infty} [2^{r\nu} \omega_k(f; 2^{-\nu})_p]^\theta \right)^{1/\theta}$.